

BRENT MATTHEW B. LACHICA

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SUMMARY

Computer Engineering graduate with hands-on experience in embedded systems, Arduino-based development, IoT solutions, and IT support. Completed an internship at MECS Solutions, assisting with technical support, hardware and software troubleshooting, and system maintenance. Currently building expertise in cloud computing through AWS Cloud Practitioner training, with a strong interest in cloud infrastructure, networking, and systems administration.

TECHNICAL SKILLS

Cloud: AWS Cloud Fundamentals, IAM, EC2, S3

Programming: C++, Python

IoT & Embedded Systems: Arduino, Sensor Integration, Environmental Monitoring Systems

Tools: Git, GitHub, Visual Studio Code

Languages: English, Filipino, Hiligaynon

Certifications:

- AWS Cloud Practitioner (Currently Pursuing)
- Cisco Networking Academy - C++ Essentials, Python Essentials, and JavaScript Essentials
- Sololearn: Introduction to C

Awards/Activities:

- Silver Medal Awardee – MSU iREX Expo 2026 (Malaysia) for the research project: "Development of an IoT-Based Wearable Device for Carbon Monoxide and Temperature Monitoring with AI-Based Alerts for Welders".
- Recognized for developing an innovative IoT and AI-assisted safety monitoring solution for industrial environments.

Organizations & Leadership:

- Member, Google Developer Student Clubs (GDSC)
- ACES Officer (Augustinian Computer Engineering Society).

WORK EXPERIENCE

IT Intern (On the Job Training) - MECS Solutions

April 2025 - June 2025

- Assisted in diagnosing and troubleshooting hardware, software, and network-related issues.
- Performed computer setup, maintenance, and system configuration tasks.
- Assisted with installation and updating of operating systems and software applications.
- Supported network connectivity troubleshooting and routine IT operations.
- Documented technical issues and solutions to improve support processes.

PROJECTS

Development of an IoT-Based Wearable Device for Carbon Monoxide and Temperature Monitoring with AI-Based Alerts for Welders

Capstone Project | 2025–2026

- Developed an IoT-based wearable device that continuously monitors carbon monoxide (CO) concentration and temperature in welding environments.
- Integrated environmental sensors with a microcontroller for real-time data collection and monitoring.
- Implemented AI-based alert mechanisms to enhance worker safety by identifying potentially hazardous conditions.
- Awarded Silver Medal at the MSU iREX Expo 2026 (Malaysia).

Solar-Powered Plant Monitoring System

Academic Project

- Designed and developed an Arduino-based environmental monitoring system powered by solar energy.
- Monitored soil moisture, temperature, and humidity levels for plant health assessment.

Arduino-Based Sanitizing Box

Academic Project

- Developed a microcontroller-based sanitizing box using Arduino technology.
- Integrated an LCD display for user interaction and system status monitoring.
- Designed the system to automate sanitization processes while ensuring ease of use.
- Applied embedded systems concepts in hardware integration and device control.

EDUCATION

Bachelor of Science in Computer Engineering

A.Y 2022- 2026

Major in Embedded Systems